

Hitaansh Jain

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Education

New York University

GPA: 3.93/4.0 | BA, Computer Science, Minor: Mathematics

- Courses: Algorithmic Problem Solving, Software Engineering, Computer Organization, Linear Algebra, Artificial Intelligence

New York, NY

Expected May 2027

Occidental College

Computer Science and Mathematics (transferred to NYU)

Los Angeles, CA

August 2024 – May 2025

Professional Experience

Header (joinheader.com)

Seattle, WA

Software Engineer Intern

May 2026 – August 2026

- Built the Follow-ups feature, letting users capture ideas from their briefings and follow them through to a recorded verdict: a 6-state lifecycle, 9 REST endpoints, 3 migrations, email reminders, and a React Native web UI, shipped to production over three weeks.
- Shipped full-stack briefing customization to production, letting users reorder, toggle, and set the detail level of each section, by refactoring a hardcoded prompt into blocks assembled from a JSONB layout model, and cut TL;DR length by 45% with an A/B-tested word budget.
- Engineered an API that lets AI coding agents drive Follow-ups concurrently with their users: row-locked, forward-only sync with an audit event per change, credential-derived verdict authority, per-key approval gating, and a compare-and-set fix for a delete race.
- Directed and reviewed Claude Code daily under hooks that blocked test rewrites and untested pushes, writing 750+ tests across an LLM pipeline, FastAPI/PostgreSQL backend, and React Native/Expo frontend, with every merge through senior review.

MathGPT (mathgpt.ai)

Remote

Software Engineer Intern, AI (Part-Time)

May 2026 – August 2026

- Designed and built a five-stage Claude pipeline (TypeScript, Next.js, MySQL) turning one calculus problem into a LaTeX case study, flashcards, and a walkthrough, with a critic that re-solves each problem.
- Imported 1,985 production templates from 3 textbooks into MySQL by writing HTML-to-LaTeX and AsciiMath-to-LaTeX converters (98% of answers converted), making linear algebra templates usable.
- Generated 900+ concept and problem flashcards, batching 622 problem decks through the Claude Batch API at about \$0.04 each, behind Zod and answer-checking gates that sent 15% back for retry before human review.
- Cut API cost per full pipeline run by 46% on average by profiling per-stage tokens (74% hidden reasoning), setting reasoning effort per stage, and running the critic's solve in parallel with generation.

Vardhman Infotech

Remote

Software Engineer Intern (Part-Time)

December 2024 – May 2025

- Met a client's offline, zero-infrastructure requirement for a Python inventory system serving 1,500+ retail locations by migrating it from MySQL to Access: a 394-column schema across 19 tables, 700+ queries across 43 files, and a 73-function database layer refactored into a metadata-driven architecture.

Projects

Local RAG Search Platform

July 2025 – Present

- Built a local RAG system (FastAPI, Ollama, ChromaDB, React) that ingests 1,070 pages of zoning codes through OCR and table-aware extraction, serving an open model in network-isolated containers at a 1.7-second median response.
- Designed a citation-grounding step that checks every cited claim against its source passage and abstains instead of answering when support is missing.
- Built BM25-weighted hybrid retrieval and an evaluation harness for retrieval, answer, and citation-support accuracy, raising page-level hit@3 from 64% with vector search alone to 89% on 28 excerpt-verified questions.

VocabLearn

February 2026 – May 2026

- Led backend on a 5-person Agile team, delivering a JWT-authenticated Node.js/Express REST API with 13 endpoints and 21 integration tests, plus Gemini 2.5 Flash with structured JSON output for reverse-dictionary search-by-meaning.

Swordfight (Unity, C#) | Gameplay & Systems Programmer

September 2023 – May 2024

- Shipped a 14-character top-down fighting game to itch.io on a 10-person student team, drawing 150k page views and 4.5 stars across 111 ratings.
- Owned the hierarchical state machine (HFSM) enemy AI and polymorphic weapon architecture behind multi-stage boss fights, and ran the PlasticSCM branch strategy that kept unmergeable Unity scenes conflict-free.

Computer Skills

Languages: Python, TypeScript, JavaScript, Java, C, C#, SQL

Frameworks: FastAPI, Flask, React, Next.js, React Native (Expo), Node.js, Express, Vite, Unity

Data & Infrastructure: PostgreSQL (Alembic, row-level locking), MySQL, MongoDB, ChromaDB, Docker Compose, GitHub Actions CI/CD, server-sent events

AI & LLM: Anthropic, Gemini, and Ollama APIs; RAG (hybrid BM25 + vector retrieval), prompt caching, Zod-validated structured outputs, LLM evaluation harnesses, OCR (PyMuPDF, RapidOCR), LangChain

Tools & Testing: Claude Code, Git, Pytest, Vitest, Mocha, Playwright